

Gabriel Larot

Software Engineer

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SUMMARY

Software engineer with experience building and evolving backend systems in production environments at scale. Focused on designing low-latency services, implementing decision logic, and maintaining reliable system behavior under real-world traffic. Comfortable working across service boundaries, handling edge cases, and improving systems through iterative development and feedback.

EXPERIENCE

Amazon

Seattle, WA

Software Development Engineer I

Mar 2025 - Present

- Owned and evolved backend services supporting dynamic content delivery, implementing selection and filtering logic operating under projected peak loads of **~160k TPS**.
- Designed backend flows that account for user context, content availability, and fallback behavior, ensuring consistent system outcomes under production traffic.
- Improved decision quality by refining selection logic, increasing successful matches by **~2%** while maintaining strict latency requirements.
- Extended service capabilities to support additional platform surfaces, coordinating cross-service changes and ensuring safe, incremental production rollouts.
- Defined and maintained API contracts and data flows for downstream consumers, prioritizing backward compatibility and operational safety.
- Diagnosed and resolved production issues related to **latency** and **service reliability** through on-call rotations, monitoring, and structured debugging.

Data Annotation

San Jose, CA (remote)

Software Validator

Feb 2024 - Jun 2024

- Reviewed and evaluated AI-generated code across **Python, Java, SQL, and HTML**, focusing on correctness, maintainability, and performance tradeoffs in production-style scenarios.
- Identified logical errors and inefficiencies, improving overall code quality through structured review processes.

Stanford University

Stanford, CA

Research Scientist, Intern

May 2022 - Sep 2022

- Modeled and simulated physical systems using **C++** and **MATLAB**, analyzing performance and stability across varying conditions.
- Wrote technical reports and presented results at internal meetings with Stanford's Plasma Dynamics Modeling Laboratory (PDML) doctoral candidates, identifying minor efficiencies in algorithmic designs.
- Evaluated tradeoffs in numerical models, improving runtime efficiency and correctness through iterative experimentation.
- Contributed software to support academic research workflows, translating experimental requirements into reliable, reusable code.

U.S. Navy

Norfolk, VA

Aviation Ordnanceman

Aug 2016 - Aug 2021

- Operated in high-reliability, safety-critical environments requiring strict procedures, detailed troubleshooting, and accountability under pressure.
- Managed **\$500,000** worth of repair parts as the Repair Parts Petty Officer (RPPO) while maintaining **100% inventory accuracy** and ensuring timely procurement to support critical divisional operations.
- Diagnosed mechanical and system-level issues through structured analysis, documentation review, and coordination with cross-functional teams to restore operational readiness.

EDUCATION

San Jose State University

San Jose, CA

Bachelor of Science, Data Science

Aug 2021 - Dec 2024

3.91 GPA

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Design, Adv. Java and Python Programming

SELECTED TECHNICAL WORK

Dynamic Contextual Ads Selection System (Amazon)

- Implemented a **backend decision pipeline** responsible for selecting relevant content under strict latency constraints.
- Designed filtering logic to handle incomplete data, invalid states, and fallback scenarios without degrading system behavior.
- Integrated historical context into selection decisions while balancing performance, correctness, and consistency.
- Extended service capabilities to support additional platform surfaces, coordinating cross-service changes and ensuring safe, incremental production rollouts.
- Validated system behavior through **staged rollouts** and **monitoring**, ensuring safe deployment under production traffic.

PROJECTS

Weather Impact on Bay Area Energy Consumption

https://github.com/gabrield03/cs163/tree/main/src_sample/interactiveWebpage

- Designed and implemented a full-stack data application to analyze and forecast regional energy demand, integrating data ingestion, modeling, and visualization into an interactive system using **Python** and **Dash**.
- Built a data processing pipeline to aggregate and clean historical weather and energy datasets, aligning time-series inputs across multiple sources to support consistent model training and evaluation.
- Evaluated forecasting approaches using both statistical and deep learning methods, implementing **SARIMA** models for interpretability and **LSTM** networks to capture nonlinear temporal patterns.
- Compared model performance using error-based metrics, achieving **~7% error** relative to baseline consumption while analyzing tradeoffs between model complexity, interpretability, and generalization.
- Applied SHAP and Partial Dependence Plot (PDP) techniques to identify key weather features influencing predictions, improving transparency of model behavior and validating feature importance.
- Structured the application to use precomputed model outputs and optimized data loading, improving responsiveness and reducing runtime overhead in the web interface.

Multithreaded Car Configuration Service

<https://github.com/gabrield03/Car-Configuration-Application>

- Designed a **thread-safe backend system** to manage shared configuration state under concurrent access.
- Implemented synchronization mechanisms (e.g., **locks/concurrent** data structures) to ensure safe state updates across threads.
- Identified and mitigated race conditions by enforcing atomic operations and controlled access patterns.
- Designed **REST APIs** for configuration updates and retrieval, ensuring consistent state across concurrent clients.
- Simulated concurrent workloads to validate system behavior under contention and evaluate tradeoffs between consistency and performance.

Recipe Application

<https://github.com/gabrield03/RecipeApplication>

- Built a full-stack recipe-sharing application with backend APIs supporting reviews, voting, and user interactions.
- Designed relational data models and optimized query patterns, reducing query latency by ~20% through indexing and efficient access strategies.
- Structured backend services to separate business logic and data access, improving maintainability and scalability.

Sudoku Web Application

<https://github.com/gabrield03/Sudoku>

- Developed a backend-driven application with stateful sessions and validation logic enforcing Sudoku constraints.
- Designed server-side validation to prevent invalid moves and ensure consistent game state across user interactions.
- Managed persistent game state through API design, supporting reliable progression and repeat interactions.

ENGINEERING APPROACH

- Design backend systems with a focus on correctness, latency, and predictable behavior under production traffic.
- Prioritize backward compatibility and incremental rollout strategies when evolving distributed services.
- Use monitoring and validation to confirm system behavior and support safe deployment.
- Approach system design through tradeoff analysis across performance, reliability, and maintainability.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript

Backend & Systems: Distributed systems, REST APIs, Service-oriented architectures

Infrastructure: AWS, Relational databases, Unix/Linux environments

Concepts: Concurrency, Low-latency system design, Performance optimization

Tools: Git, CI/CD pipelines, Unit & integration testing